Jonathan Yu

jonathandyyu@berkeley.edu | in/jonathandyyu | (408)-335-9445

Education

University of California, Berkeley Expected May 2028

B.S. Chemistry, B.A. Computer Science; GPA: 3.90

Relevant Courses: Data/Program Structures, Linear Algebra, Discrete Math, Probability, Data Science Foundations, Organic Chemistry

Relevant Experience

Researcher–Gong Lab January 2025 – Present

UC Berkeley, School of Optometry–Berkeley, CA

- Implemented doublet filtering, clustering, and pseudotime workflows in Seurat to identify age- and genotype-specific cell subtypes
- Analyzed regulatory/transcriptional dynamics in mouse lens development across 1,000+ genes and 800+ transcription factors
- Profiled differentially expressed genes with gene ontology analysis to identify candidate regulators in lens growth and homeostasis
- Maintained 1,000+ mice with critical genotype modifications; extracted lens tissue for proteomic and transcriptomic profiling

iGEM Research and Finance Intern Director January 2025 – Present

UC Berkeley, Department of Bioengineering–Berkeley, CA

- Identified 1NVU Ras.GTP as a protein-protein interaction target optimized for proposed *de novo* peptide inhibitor design pipeline
- Evaluated 30+ candidates generated by RFDiffusion using ProteinMPNN sequence design and AlphaFold PAE/pLDDT scoring
- Optimized 5 partial diffusion parameter values to iteratively refine and introduce structural diversity for peptide candidates
- Sourced $2,000+ in funding and hosted 400+ attendees across 10+ industry & academic partners for Bay Area BioE Symposium

Rockefeller SSRP Intern, Fuchs Lab June 2024 – August 2024

The Rockefeller University – New York, NY

- Designed and performed cell stressor/cytoprotector perturbation protocol for keratinocytes and squamous cell carcinomas
- Quantified morphological and proliferative effects using immunofluorescence imaging and segmentation pipelines in ImageJ
- Achieved 40% decrease in cell proliferation across psoriatic and unfolded protein responses conditions in presence of retinoic acid
- Identified perturbation-specific transcriptional signatures in epithelial Bulk RNA-seq datasets with DEG and GSEA analysis in R
- Presented at Rockefeller Summer Poster Session; highlighted regulatory mechanisms in skin barrier maintenance

UCSC SIP Intern, MacMillan Lab June 2023 – August 2023

UC Santa Cruz, Department of Chemistry and Biochemistry – Santa Cruz, CA

- Cultured and extracted novel non-enzymatic natural products from *Erythrobacter* primed with fluorinated anthranilic acid scaffold
- Developed and executed analytical methods to purify 20+ hits using iterative flash chromatography, HPLC, and 1H/19F NMR
- Interpreted LCMS fragmentation patterns to infer reactivity and structural motifs; validated structure via COSY/HSQC/HMBC NMR
- Presented at UCSC SIP Research Session, showcasing reactivity-based natural product synthesis with one final lead molecule

SSP Participant at Indiana University June 2022 – July 2022

The Summer Science Program – Bloomington, IL

- Identified PdCdc14 as a key target in preventing citrus crop rot by inhibiting its mitotic phosphatase function in *P. Digitatum*
- Produced and extracted PdCdc14 via recombinant protein expression and Ni-NTA column resin purification
- Designed and performed enzymatic assays to quantify catalytic activity, substrate specificity, and inhibitor IC50 values
- Computationally generated homologous PdCdc14 scaffolds; verified ortholog structure conservation using sequence alignment
- Scored docking affinities of 12 FDA-approved small-molecule scaffolds and 40+ respective derivative candidates in CCG MOE
- Submitted findings on catalytic kinetics, inhibitor candidates, and active site sterics to *Purdue University Research Repository*

Skills, Interests, and Involvements

Programming & Analysis: Python, R, Java, bash scripting, HPC, biopython, Scanpy, pandas, Seurat
Modeling & Tools: ImageJ, RFDiffusion, AlphaFold, PyMOL, SnapGene, MNova, Xcalibur
Wet-Lab Techniques: NMR, LCMS, GCMS, HPLC, Protein Purification, Cell Culture, Enzyme Assays, PCR, Western Blot
Mentorship and Teaching: Berkeley CS Mentors Junior Mentor for Data C88C; SSP Connect mentor for SSP Alum
Interests: bioML, traveling, transportation networks, hiking, fragrance, floriculture, scrapbooking, zinemaking, board games